

신호 및 시스템(공) (EE301-01)

학기	2026년 1학기
담당교수	최재호
이수구분	교선�필수
학점	3.0
강의시간	화09:00-10:30(E7 - 241), 목09:00-10:30(E7 - 241)
강의실	E3 414 (교수실)
소속	기초학부

교과개요

In this course, students will learn how to understand signal (e.g., sound, radio waves, and electrical impulses) behavior in a mathematical way and how to analyze/design systems that process those signals. This course provides a foundation for various areas related with electrical engineering, such as signal processing, communications, and control theory. We will also implement various functions to analyze signals and systems through Python.

교과목표

1) Understand fundamental concepts of signals and systems, 2) Learn how to analyze signals in both time and frequency domains, 3) Understand LTI systems and their properties, and 4) Get familiar with mathematical tools like Fourier transform and Laplace transform.

주차별 강의계획

1주차:Introduction to Signals
2주차:Introduction to Systems
3주차:Discrete-Time Linear Time Invariant Systems
4주차:Continuous-Time Linear Time Invariant Systems
5주차:Continuous-Time Fourier Series
6주차:Continuous-Time Fourier Transforms
7주차:Applications of Fourier Transforms
8주차:Mid-Term Exam
9주차:Discrete-Time Fourier Transform
10주차:Sampling
11주차:Laplace Transforms I
12주차:Laplace Transforms II
13주차:System Design and Analysis
14주차:z-Transforms I
15주차:z-Transforms II
16주차:Final Exam

학습윤리

No cheating; Use of phones during exams is strictly prohibited.

교과정책(방침)

Lecture-based course; Hands-on coding projects in Python.